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Mirror stabilization experiments in the Hanbit device

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Abstract

The Hanbit magnetic mirror device has been involved in a series of experiments on stabilization of the MHD flute type mode. We undertook investigations to see whether a divertor and the kinetic stabilizer (KS) of Post *et al* can stabilize the MHD instability. The Hanbit divertor configuration used one of the central cell coils with reversed current as the divertor coil and two other coils with increased current to compensate the field droop. The divertor eliminated the $m = -1$ instability when the null point was inside the vacuum tank. The KS used microwave produced plasmas on field lines in the cusp tank region. Two coils on the cusp tank were configured to produce expanding field lines with a large positive radius of curvature. A 5 kW 2.45 GHz magnetron was used to produce an electron cyclotron resonance heated plasma pressure in this region. A reduction in instability duration was observed for high power plasmas. However, for low power plasmas that terminate violently with an $m = -1$ instability, the KS action made the duration of the instability longer.

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(Some figures in this article are in colour only in the electronic version)

1. Introduction

Hanbit [1] is about half of the original TARA [2] tandem mirror device from MIT. Earlier work in Hanbit [3] showed that it was possible to stabilize the $m = -1$ flute type MHD instability with RF power near the cyclotron resonance by the sideband coupling process. We undertook investigations to see whether the $m = -1$ MHD flute-like instability could be stabilized by a mirror divertor [4–7] and the kinetic stabilizer (KS) of Post *et al* [8–10].

According to the simple theory [4], a divertor stabilizes the plasma by providing a region of low poloidal field at a separatrix in which the electrons can drift azimuthally. The high plasma conductivity causes the potential of perturbations with azimuthal numbers $|m| > 0$ to vanish and thus shorts out the electric field due to the interchange instability. There is no sheath, so it is equivalent to an electron emitting end wall. In order for the separatrix to be effective, it must be in the vacuum chamber and connected to field lines at a smaller radius than the limiter. On the other hand, according to Pastukhov [11] the main stabilizing effect is compressibility. High mode numbers

should be stabilized by finite Larmor radius effects. Under the influence of the compressibility, sufficiently gradually decreasing (marginally stable) pressure profiles become stable for all interchange flute-like modes. The simplest one fluid isotropic pressure model leads to a marginally stable (MS) pressure profile for the flute-like modes of the form

$$S = pU^\gamma = \text{const}, \quad U(\psi) = \int_{-}^{+} dl/B,$$

where S is a single-valued function of the entropy density, p is the pressure, U is the specific flux tube volume, $\gamma = 5/3$ is the adiabatic index and l is the distance along the field line. This means that the radial pressure profile is of the form

$$p \propto U^{-5/3}.$$

Because the fields are axisymmetric, there is no deterioration of radial confinement as in conventional tandem mirrors with minimum B anchors. Non-axisymmetric anchor fields can lead to the deterioration of radial confinement.

Post *et al* [8–10] proposed the KS to stabilize the MHD instabilities in mirror devices. This idea was based on the

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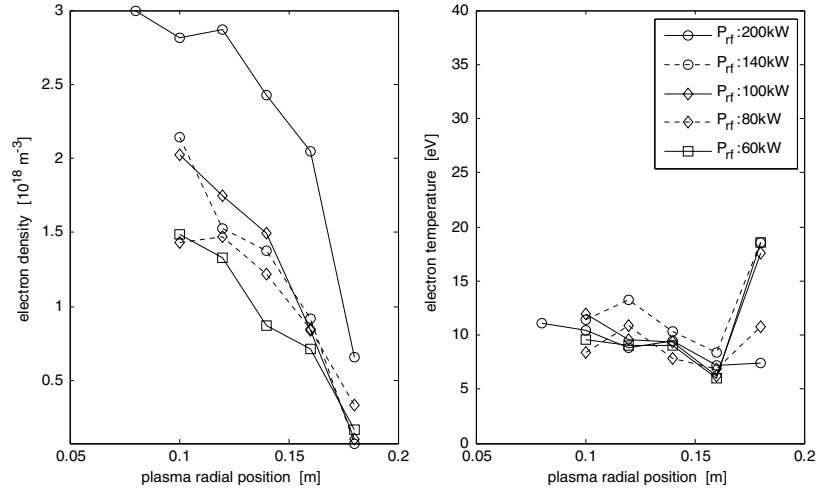


Figure 1. Electron density and temperature profiles. The data points were obtained by a rapidly scanning Langmuir probe.

concept of Ryutov [12] derived from the work of Rosenbluth and Longmire [13]. The stability of the gas dynamic trap [14] has been attributed to this concept. The KS concept involves locating a stabilizing plasma pressure on the field lines at a critical region in the expanding field region with good field line curvature outside the mirrors so that the main plasma in the mirror central cell in regions of unfavourable curvature can be stabilized. If the stability integral, I [12, 13] is positive, the plasma will be stable. The stability condition is

$$I = \int_{-L}^{+L} a^3 \frac{d^2 a}{dz^2} [p_{\perp} + p_{\parallel} + \rho v^2] dz > 0,$$

where a is the outermost radius of the plasma, z is the axial distance, $p_{\parallel}$ is the parallel plasma pressure, $p_{\perp}$ is the perpendicular plasma pressure, v is the flow velocity, ρ is the plasma mass density ($n_i m_i$) where the ion density is n_i and the ion mass is m_i . The integration is to be done from one end ($-L$) to the other end ($+L$) of the entire region containing plasma. Post's proposal [8] was to use low energy (e.g. ~ 1 keV) high Z (Cs^+ or Xe^+) ion beams to be injected along field lines in the expanding field region at the end of a mirror machine. We were unable to acquire such ion beams.

In section 2 we describe the Hanbit device and a description of the diagnostics. We discuss the earlier divertor experiments in TARA and HIEI in section 3. In section 4 we discuss the divertor experiments in Hanbit. A discussion of ion loss from the magnetic null and the ability of the probes to detect the instability even if they are outside the separatrix is given in section 5. A detailed probe data analysis is found in section 6. The design and earlier measurements on the KS plasma in the cusp region of Hanbit are discussed in section 7. In section 8 we discuss the most recent KS experiments in Hanbit including analysis of the probe data. A discussion in section 9 is followed by a summary and conclusions in section 10.

2. The Hanbit device and diagnostics

As mentioned above, Hanbit is a tandem mirror device formed from components of the original TARA device. It has a central

cell, one anchor with a fan out tank, one plug and a cusp tank. The central cell is ~ 5 m in length and has a midplane plasma radius of 0.186 m determined by limiters. The field in the central cell is ~ 0.24 T and the mirror ratio is typically 10. For all the experiments described here the anchor coils were turned off so as not to influence the stability. The plasma was formed with power supplied to a slot antenna by a 3.5 MHz transmitter. Normally a power of 200 kW was used but the KS experiments used 200 kW and later 80 kW in order to produce more unstable plasmas.

The main Hanbit diagnostics were diamagnetic coils, a single 94 GHz interferometer, a fast insertion probe and several probe arrays mounted on limiters. For the divertor experiments the diamagnetic coils could not be used due to the necessity of producing linear excursions (ramping) of the magnetic fields during the plasma discharges. They were of very limited use during the KS experiments because the signals were so small. The main probe diagnostics were two probe arrays each having four probes mounted at 90° intervals on limiters on opposite ends of the central cell.

The electron temperature and density have been measured by Langmuir probes in previous experiments [15] with very similar conditions. Typically, the electron density has a peak of $\sim 3 \times 10^{18} \text{ m}^{-3}$ and the electron temperature has a central value of ~ 10 eV with a flat profile for an ICRH power of 200 kW. The density is in agreement with the interferometer. Repeated attempts have been made to measure the ion temperature under these conditions without success both due to the low temperature and low density. Some measurements [16] have been made at much higher neutral gas pressure than for these experiments and hence are not relevant. The neutral pressure in the central cell for these experiments at the end of the RF pulse was typically $(1-3) \times 10^{-6}$ Torr corresponding to a hydrogen gas density of $\sim 3 \times 10^{16}$ to $\sim 1 \times 10^{17} \text{ m}^{-3}$.

Figure 1 shows measurements of electron temperature and density under similar conditions to those used for these experiments. The data taken at the axis are not reliable and are not plotted. The limiters are located at 0.186 m and the electron temperature and density at 0.18 m are not reliable due to the low ion saturation current.

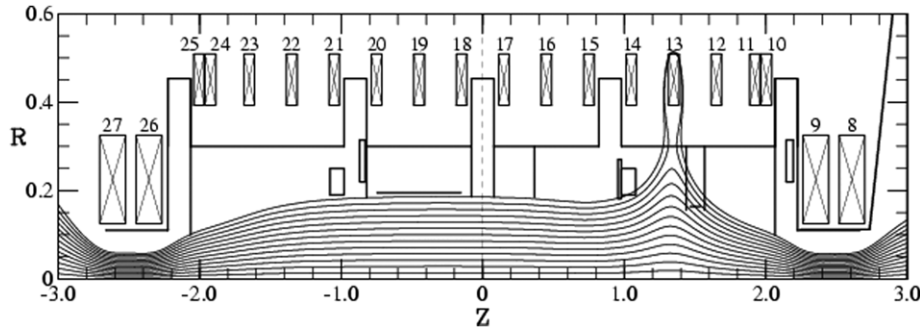


Figure 2. Far west end divertor using coil 13 plus coils 14 and 15 energized to strengthen field under the SLOT antenna for better matching. The horizontal SLOT antenna is shown just to the left of the dotted vertical line at the midplane under coils 18–20.

3. Earlier divertor experiments on TARA and HIEI

A TARA tandem mirror divertor [4] was designed and built at MIT. Subsequent pioneering experiments [5] showed that the stability boundary was extended by the use of this divertor. The stability was enhanced by mapping the null radially into the plasma. According to Pastukov [11], due to the nature of the peaked plasma profile in TARA and the design of the divertor, it was not possible to fully stabilize the TARA plasmas. Yasaka *et al* designed two divertors for the HIEI device [6] and found that the design that provided smaller radius nulls produced the best stabilization. Subsequent investigations [7] showed that stabilization was obtained for the null inside the limiter and that the density gradient was much larger for the divertor stabilized plasmas.

4. Hanbit divertor experiments

We could not modify the central cell tank or add additional coils, so various central cell coils were reversed and used with a separate power supply as divertor coils. Experiments performed with a symmetric double divertor showed that it was not possible to establish plasmas with it. Although an asymmetric divertor will cause enhanced neoclassical resonant transport [17], it can still stabilize the flute mode. A single divertor powered by a separate supply in one end was chosen for the main experiments. Two adjacent coils were also powered by another supply to increase the field and prevent field droop due to the divertor coil. The diverted plasma was directed along field lines to the wall near the divertor coil and there was no possibility of removal of impurities or auxiliary pumping of the diverted plasma. Figure 2 shows the field lines and $|B|$ contours for a representative set of currents for the chosen configuration. The coils are numbered in sequence from right to left starting with 8 and ending with 27. Coil 13 is used as the divertor coil and coils 14 and 15 have increased current to compensate the field droop due to the divertor coil.

Figure 3 shows the field lines and $|B|$ contours for the region in the vicinity of the wall under the divertor coil. The divertor current is -190% where 106% is the normal current for ICRH heating in the central cell. Optimum heating occurs for currents from 106% to 110% . The field null point (x-point) is at the wall. This is the current at which the divertor first stabilizes the $m = -1$ flute-like instability. More negative currents also stabilize it and less negative currents have no effect on the instability. The instability limit is very abrupt.

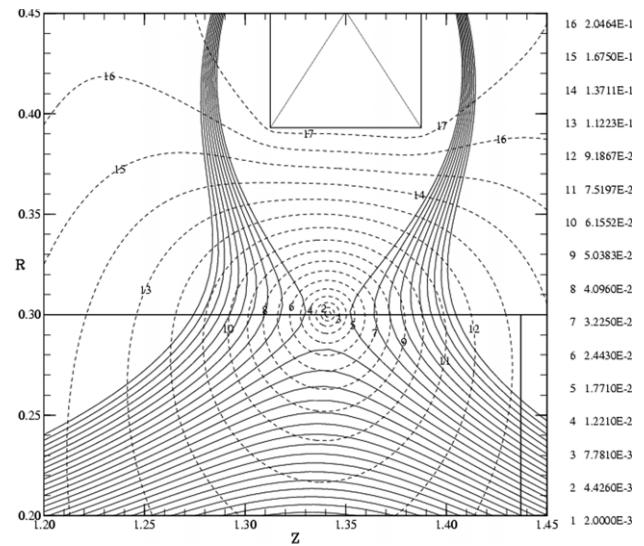


Figure 3. Close up view of the divertor region for the west end divertor as in figure 2. The $|B|$ contours are shown as dashed lines. For a divertor current of -190% , the calculated x-point is at the wall.

Even with a single asymmetric divertor in the end far from the antenna, ramping on the divertor caused frequent ICRH power trips. It was not possible to establish the plasma with the divertor fully on. Normally the plasma wall conditioning was done with all the central cell (CC) currents 110% of normal and then switched to a configuration with the divertor coil 13 with zero current (0%) and the compensation coils 14 and 15 with currents 170% of normal. This arrangement meant that the field under the SLOT antenna (i.e. under coils 18–20) was $\sim 110\%$. After conditioning, it was possible to ramp on the divertor coil current from an initial value between 0% and -100% to a divertor current as negative as -230% . For these experiments we normally produced plasmas with 200 kW of ICRH power. Without a divertor, these plasmas were stable against the interchange mode until the RF power ramp down at the end but always became unstable at low density near the end of the ICRH ramp. If the ICRH power tripped during a shot, the $m = -1$ signal at the end of the shot was rarely seen. A smooth ramp down to zero power was necessary to produce the $m = -1$ instability for plasmas created with a power of 200 kW . In our study we excluded all shots with a power trip. These experiments have also been briefly described in a recent publication [18]. During these experiments benign

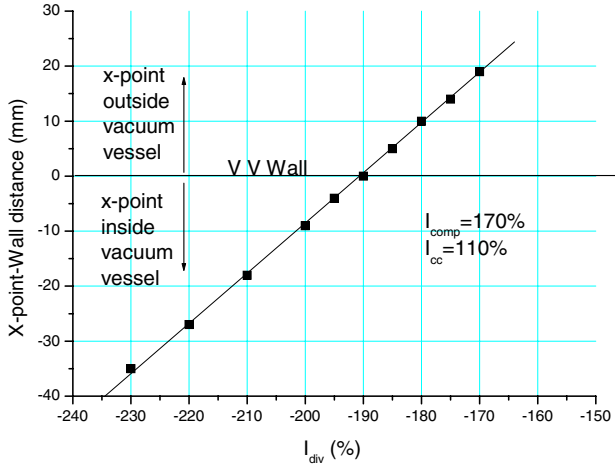


Figure 4. Location of the x-point as a function of the divertor current with the CC current and compensation currents held fixed. The distance from the wall varies linearly with the divertor current.

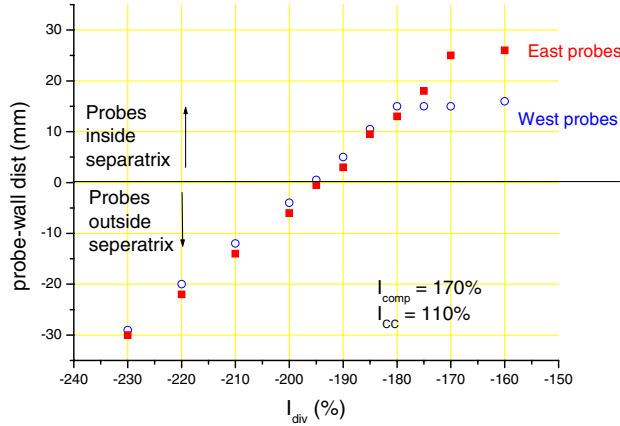


Figure 5. Probe-plasma edge distance as a function of the divertor current. The east probe locations are given by closed squares. The west probe locations are given by open circles. The probes are far inside the plasma for low divertor currents. As the divertor current increases the probes leave the plasma and are outside the separatrix. At a divertor current of -190% , both probe arrays are inside the separatrix. For divertor currents of -195% both probe arrays are at the separatrix and for -200% and higher currents both probe arrays are outside the separatrix.

drift waves are usually observed instead of flute modes during the ramp down. The drift wave can be transformed into the flute mode by gas pressure and power changes and both modes were extensively studied and identified earlier under very similar conditions to the present experiment [19]. Line tying does not appear to be a factor because the plasma is always in contact with the wall for sufficiently large negative divertor currents ($I_{\text{div}} < -160\%$) and stability is found only when the x-point is inside the vacuum wall.

As noted in figure 3, the analysis showed that the x-point was at the wall when the stability was first observed. Figure 4 shows the x-point location relative to the vacuum vessel wall as a function of the divertor current. The calculated locations are estimated to be accurate within about 2 mm. The distance from the wall varies linearly with the divertor current.

Figure 5 shows the location of the east and west probe arrays as a function of the divertor current with the CC and

compensation coil current held fixed. When the divertor current is low the probes are far inside the separatrix and as the divertor current increases the probes approach the separatrix and finally go outside the separatrix for sufficiently high divertor currents. The calculated locations are estimated to be accurate within about 2 mm.

It appears that the probes were at and outside the separatrix during the period when the divertor was turned on sufficiently to produce stability. Hence, it appears that the probes may not have been able to detect the plasma during the instability. However, this is not the case as will be discussed in section 5.

5. Ion loss through the magnetic null region

Flute instabilities cause ion radial loss [20]. However, the un-magnetized electrons around the magnetic null region can move freely in the azimuthal direction, so that the flute modes are expected to be stabilized by the short circuit at the separatrix. Therefore, ions do not escape to the outside of the separatrix by flute instabilities.

A divertor in an open system such as a mirror device is different from a scrape-off layer in a closed system such as a tokamak. The electrostatic fluctuations of flute modes inside the separatrix do not spread beyond the separatrix. So this section clarifies the process as to how a probe outside the separatrix picks up the signal from inside the separatrix.

The equation of ion motion is described by the pseudo-potential Φ , because the ion angular momentum around the axis is conserved in an axisymmetric magnetic system such as Hanbit, where

$$\ddot{r} = -\frac{\partial \Phi}{\partial r}, \quad \ddot{z} = -\frac{\partial \psi}{\partial z}, \quad r\dot{\theta} = -\frac{q(\psi - \psi_0)}{(m_i c r)} \quad (1)$$

The symbol $\ddot{r}$ means the derivative of r with respect to time. Equation (1) can be applied to the un-magnetized ions around the magnetic null region. Here

$$\Phi(r, z) \equiv \frac{q^2}{2m_i^2 c^2 r} (\psi(r, z) - \psi_0), \quad (2)$$

$$\psi(r, z) \equiv \int_0^r B_z(r', z) r' dr'.$$

Figure 6 shows a plot of the equi-contour surface of $\Phi(r, z)$ which gives the region inside in which a 10 eV ion can move freely. The equi-contour surface includes the magnetic null point inside it, that is, the ions inside the equi-potential surface are non-adiabatic and their magnetic moment μ is not conserved.

Figures 6(c) and (d) plot the equi-potential surfaces in figure 6(b), respectively. The east and west probe tips shown by the bold rectangles in figures 6(a) and (b) are outside the equi-potential surfaces in figure 6(c) in the case of $I_{\text{div}} = -200\%$, while those probes are inside the equi-potential surface in the case of $I_{\text{div}} = -195\%$.

As mentioned above, ions inside the equi-potential surface in figure 6(b) have an isotropic distribution because the conservation of magnetic moment is broken during one bounce motion along a magnetic field line. The ion density $n(l)$ with an isotropic distribution is constant along a magnetic field line.

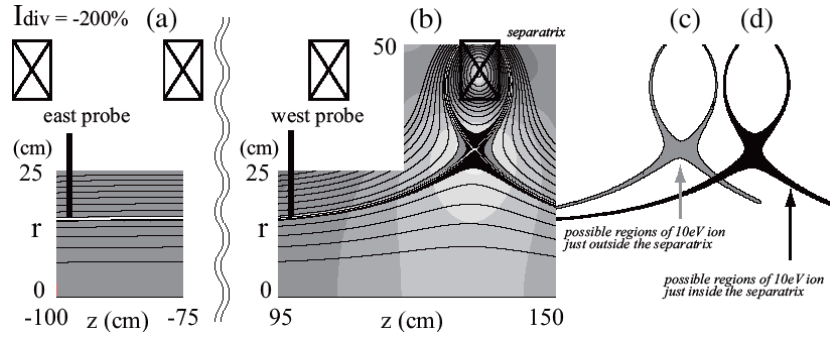


Figure 6. Contour plot of the pseudo-potential of a 10 eV ion (c) just outside the separatrix and (d) just inside the separatrix for the case of $I_{div} = -200\%$.

On the other hand, the number of ions is calculated by the equation $N = \int n(l)dl/B(l)$ where the integration is carried out along a magnetic field line. That is, the local number of ions is proportional to $n(l)/B(l)$, which means that the ion population around the magnetic null region is much larger than it is in the other regions.

As long as the angular momentum of the ions is conserved, the transfer of ions in the region of figure 6(d) to that in figure 6(c) is inhibited, although both regions overlap around the magnetic null region. Small Coulomb collisions, however, can break the conservation of angular momentum which exchanges the ions in the region of figure 6(d) with those in the region of figure 6(c).

Noting that the probability of existence of the ions is proportional to the local particle number of ions, the ions near the separatrix have a larger probability of receiving collisions near the magnetic null region than the other region. So ions inside the separatrix in figure 6(d) can transfer to the region outside the separatrix in figure 6(c). This is one mechanism of ions escaping from inside the separatrix to the outside when flute instabilities carry ions in the core region to the neighbourhood of the separatrix. Therefore, the probe tips outside the separatrix can detect the ion current caused by flute instabilities at least in the case of $I_{div} = -195\%$.

6. Probe data analysis for the divertor experiment

There was a probe array on each end of Hanbit with four probes so arranged that there were two probes on a field line 90° apart in four azimuthal locations, the same system as used in previous studies on Hanbit by Jhang *et al* [3] and Yoon *et al* [19]. The same checks were made as for those experiments. For all the signals we studied that had $m = -1$ rotation, the signals on the two end probes on each field line were simultaneous with an experimental error of about $10 \mu s$. This demonstrated that the wave under study had $k_{||} \sim 0$ which was consistent with the flute mode. For the KS experiments in section 8, this same instability was also studied and a clear demonstration of $k_{||} \sim 0$ is found in figures 19 and 20.

Several signals are displayed in figure 8 for shot 24476 which is a typical plasma shot with the divertor on at -160% but the x-point outside the vacuum vessel. The top frame shows the line density, the middle frame the density fluctuation from a typical probe and the bottom frame the probe signal divided by the line density and normalized to 1 for the first 10 ms.

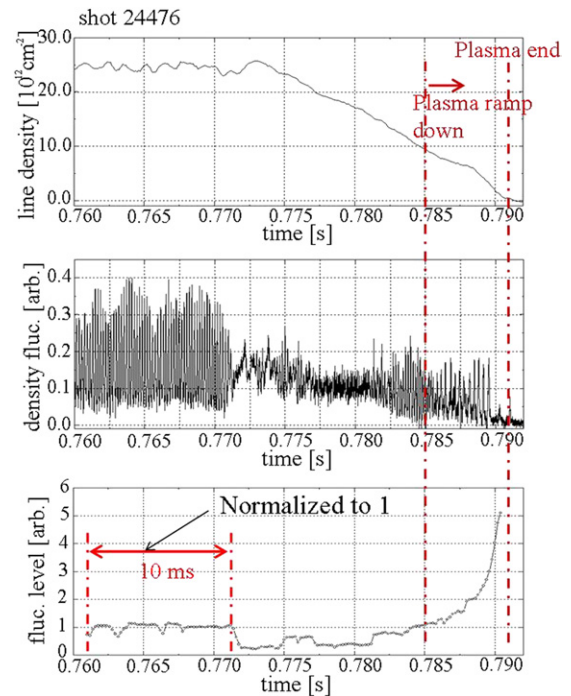


Figure 7. Analysis for shot 24476. Top frame: line density; middle frame: density fluctuations from a probe; bottom frame: density fluctuation divided by line density. The fluctuation level increases dramatically in a flute mode as the plasma terminates.

It is clear that as the plasma density ramps down, the probe signal first decreases and then dramatically increases near the end of the plasma. The signal after ~ 0.785 s is a flute mode. While the line density measured by the interferometer starts to decrease at about 0.775 s, the density as determined by probes at the edge starts to decrease at 0.785 s.

Figure 8(a) shows the signals from an azimuthal probe array on one of the limiters near the end of the ramp down for the same shot as in figure 7. The chronological sequence indicated by the dotted lines shows that these signals result from an $m = -1$ perturbation at the surface of the plasma. This is the normal sequence at the end of a shot using 200 kW of ICRH power. For this shot the divertor current was ramped on to a value of -160% in which case the x-point is outside the vacuum vessel. Figure 8(b) shows the same probe signals resulting from ramping on the divertor current to -190%

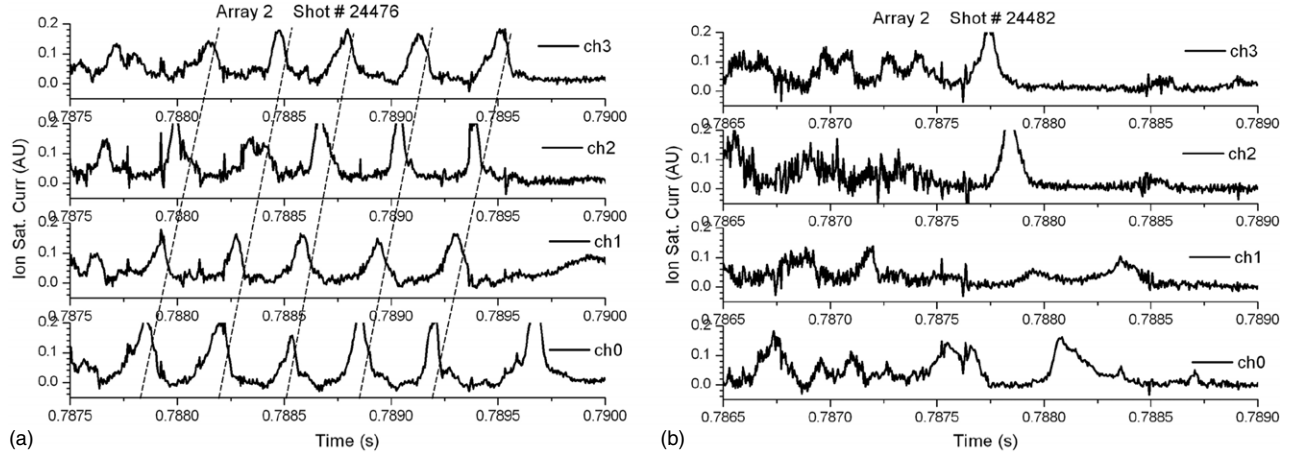


Figure 8. Signals from array 2 probes. (a) Signals with the divertor on but the x-point far outside the vacuum vessel. (b) Signals with the divertor on and the x-point at the vacuum vessel wall.

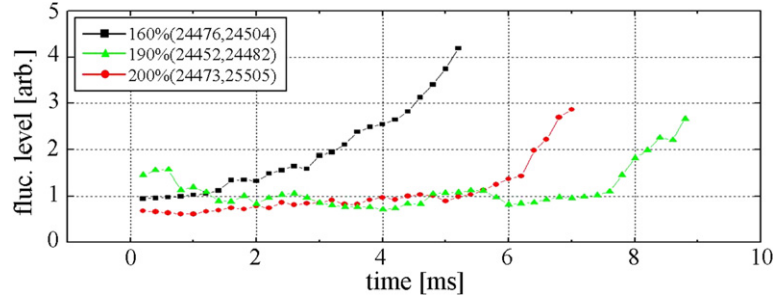


Figure 9. Fluctuation level with three different levels of divertor current. As the divertor current is increased, the fluctuation level remains lower for a longer time.

in which case the x-point is near the wall of the vacuum vessel. This is precisely the condition shown in figures 2 and 3.

For figure 8(b) the sequence of signals is difficult to determine and is not the sequence for $m = -1$. We interpret this as MHD stability. A series of experiments showed that the divertor current ramped on to -190% or more negative would produce stability but less negative currents would not remove the $m = -1$ signature at the plasma termination. The calculations showed that the x-point is outside the vacuum vessel wall for a divertor current of less than -190% . The current must be more negative than this because, according to the simple theory, not only must the x-point be inside the vacuum vessel but the magnetic field at the wall has to be high enough to contain the electrons near the null that provide the stabilization. Since we observe stabilization at -190% , we think that the x-point is actually inside the vessel for this current.

Figure 9 compares the fluctuation levels for three cases of the divertor current. The origin of the abscissa is the start of the plasma ramp down. The plasma fluctuations change depending on the divertor current. It is clearly seen that the plasma remains more stable, i.e. it has a lower fluctuation level, for larger divertor currents indicating the effect of the divertor stabilization. The initial rotation velocity due to the drift mode remains unchanged for large divertor currents while for small currents the rotation velocity changes direction as the flute develops.

There is a possibility that the difference in fluctuation level in figure 9 is caused by the difference in the RF-stabilizing effect in each shot. Figure 10 shows the RF power supplied per plasma particle versus time. The RF power levels are almost the same except for the period after ~ 6 ms for -190% shot. But the increase shown should give a larger RF-stabilizing effect.

We assume that the X - Y coordinates of the centroid of the plasma column are expressed by the equations:

$$Y = \frac{f_0 - f_2}{f_0 + f_2}, \quad X = \frac{f_1 - f_3}{f_1 + f_3}.$$

In the above equations, f_i is the signal strength detected by the probe of channel i . There were four probes used: ch0, ch1, ch2 and ch3. Figure 11 illustrates the terminology of the data analysis:

The values of X and Y are in the range of -1 to 1 , but the range may be over-scaled by noise or signal offsets. We apply this method to shot 24476 (divertor $I = -160\%$), shot 24424 (divertor $I = -190\%$) and shot 24497 (divertor $I = -195\%$). For these divertor currents, the following three figures each contain three graphs showing (top right) the time variation (from bottom to top) of the position of the plasma centroid, (X , Y), connected by a red line, (bottom right) an x - y plot as if looking down from the top, and (bottom left) an x - y plot of X and Y versus time.

Figure 12 shows the data analysis for shot 24476 with a divertor current of -160% such that the x-point is outside the

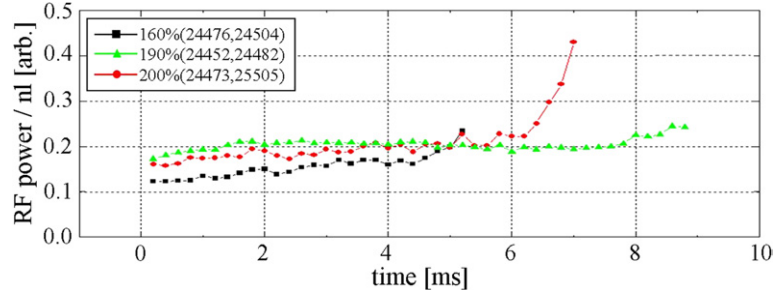


Figure 10. RF power per plasma particle for the same three shots as in figure 9. The power levels are almost the same except after ~ 6 ms in the -190% shot. This increased power should provide better RF stabilization.

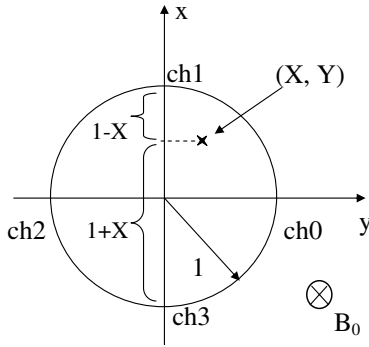


Figure 11. The terminology used for the probe data analysis.

vacuum vessel. In the lower right diagram, the dark circle is the plasma boundary and the dotted circle is the noise limit. In this case there should be no divertor action. The arrows in the bottom right graph show that the motion is counter clock wise (CCW) and hence is a flute-like mode.

Figure 13 shows the data analysis for shot 24424 with a divertor current of -190% . In this case the x-point is near the vacuum vessel wall and there can be divertor action. The arrow in the bottom right graph is not clear but appears to show a mixture of CCW and clockwise (CW) motions corresponding to flute- and drift-like modes, respectively.

Figure 14 shows the data analysis for shot 24497 with a divertor current of -195% . In this case the x-point is inside the vacuum vessel and there should be divertor action. As expected, the $m = -1$ flute-like mode is absent. The motion shown in the bottom right graph is difficult to determine.

As seen in figures 13 and 14, the action of the divertor appears to eliminate the flute-like mode. In figure 13 the drift-wave-like mode which existed earlier in the discharge continues and the flute-like mode never appears. A critical point to determine whether or not the mode is a flute mode or an electron drift mode is the direction of the phase velocity. For the flute mode, the instability is driven by ion dynamics and the wave propagates in the direction of the ion diamagnetic velocity which is $m = -1$ in the Hanbit magnetic configuration with the same sign of the radial density gradient, dn/dr . On the other hand, the propagation direction for the electron drift wave is in the electron diamagnetic direction which is the $m = +1$ direction in Hanbit. We interpret the probe measurements as indicating phase velocity and not as rotation velocity. We think that the $\vec{E} \times \vec{B}$ rotation direction will be dependent on the

direction of dn/dr and the sign of dn/dr is the same for both drift and flute modes.

We do not have any measurements of the electrostatic potential so we cannot directly calculate the $\vec{E} \times \vec{B}$ velocity and rotation frequency. Assuming the plasma potential is negative, the electric field is radially outward. This means that the $\vec{E} \times \vec{B}$ rotation direction is the same as the ion diamagnetic drift direction. So it is important to be able to distinguish between an $m = -1$ MHD flute and the $\vec{E} \times \vec{B}$ rotation.

From [21, equation 18.60], we find the poloidal $\vec{E} \times \vec{B}$ velocity given by

$$v_{\theta}^{\text{dr}} \approx (3kT_e)/(e\lambda_{T_e}B),$$

where the electron temperature is kT_e (eV), the T_e scale length ($T_e/(dT_e/dr)$) is λ_{T_e} (m) and the magnetic field is B (T).

An estimate of λ_{T_e} from figure 1 for 200 kW could be larger than the plasma radius, but the lower limit would be about 0.1 m. If we take $T_e \sim 10$ eV and $\lambda_{T_e} \sim 0.1$ m from figure 1, and $B \sim 0.24$ T, then $v_{\theta}^{\text{dr}} \sim 1250$ m s $^{-1}$. For a limiter defined plasma radius of 0.186 m, the poloidal $\vec{E} \times \vec{B}$ frequency would be ~ 1 kHz. This is smaller than both the measured drift frequency of ~ 6 kHz and the measured flute frequency of ~ 3 kHz and would be even smaller if λ_{T_e} is larger. So it appears that the $\vec{E} \times \vec{B}$ rotation frequency is too small to be mistaken for an $m = -1$ flute. In addition, whenever the $m = -1$ signal is observed, the plasma dies violently while an $\vec{E} \times \vec{B}$ rotation would not be expected to cause this effect.

As already mentioned, there are no measurements of the ion temperature under the conditions of this experiment. The density scale length at the plasma surface is about 0.01 m [15] and the magnetic field is typically about 0.24 T. Based on the measured value of the flute frequency of ~ 3 kHz, the ion temperature calculated from [22] appears to be ~ 10 eV.

The top part of figure 15 shows the left- and right-rotating components of the plasma rotation for shot 24505 with a divertor current of -160% when the x-point is outside the vacuum vessel. The bottom part of the figure shows the frequencies of the modes. For the top part of the figure we used the normalizing condition:

$$[\text{Right rotating component}]^2 + [\text{Left rotating component}]^2 = 1.$$

As seen in the top part of the figure, the right rotation changes into the left rotation after the start of the plasma ramp down. As seen in the bottom part of the figure, the drift mode

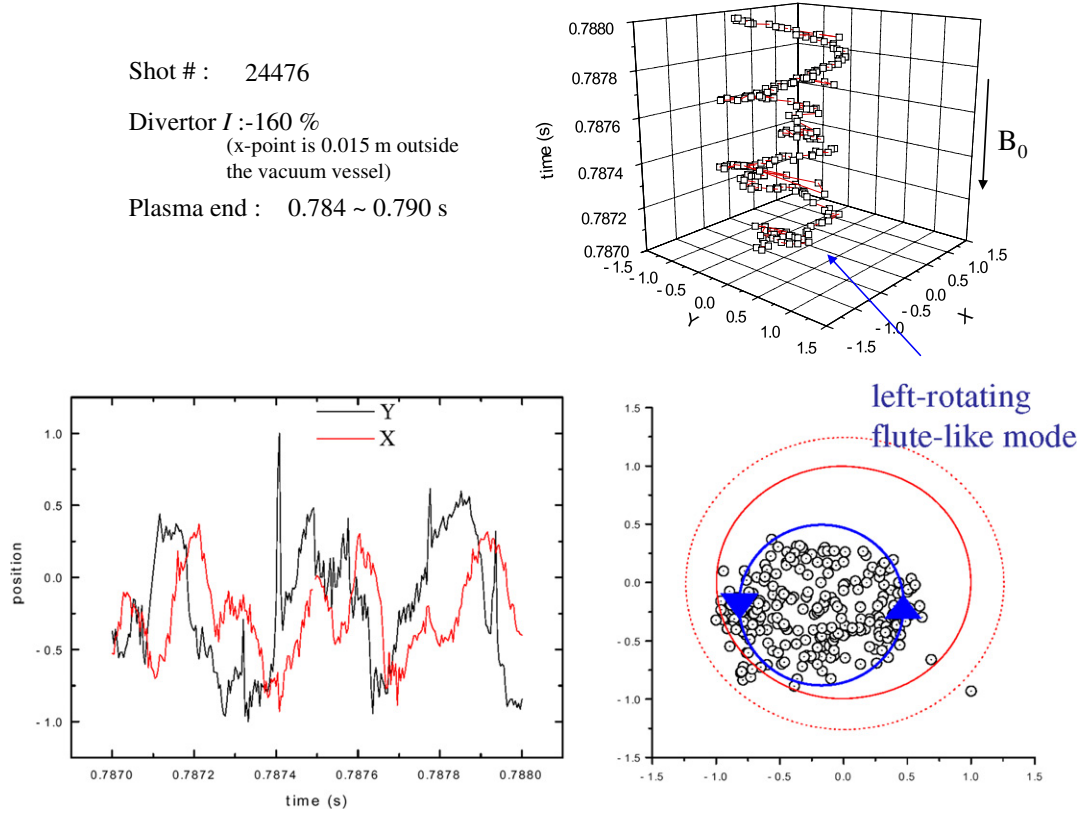


Figure 12. Shot 24476 with a divertor current of -160% . The top right frame shows a 2D view of the centroid motion versus time. The bottom right shows a view of this from the top. The bottom left shows plots of X and Y versus time. The motion is a left-rotating flute-like mode. The motion is CCW.

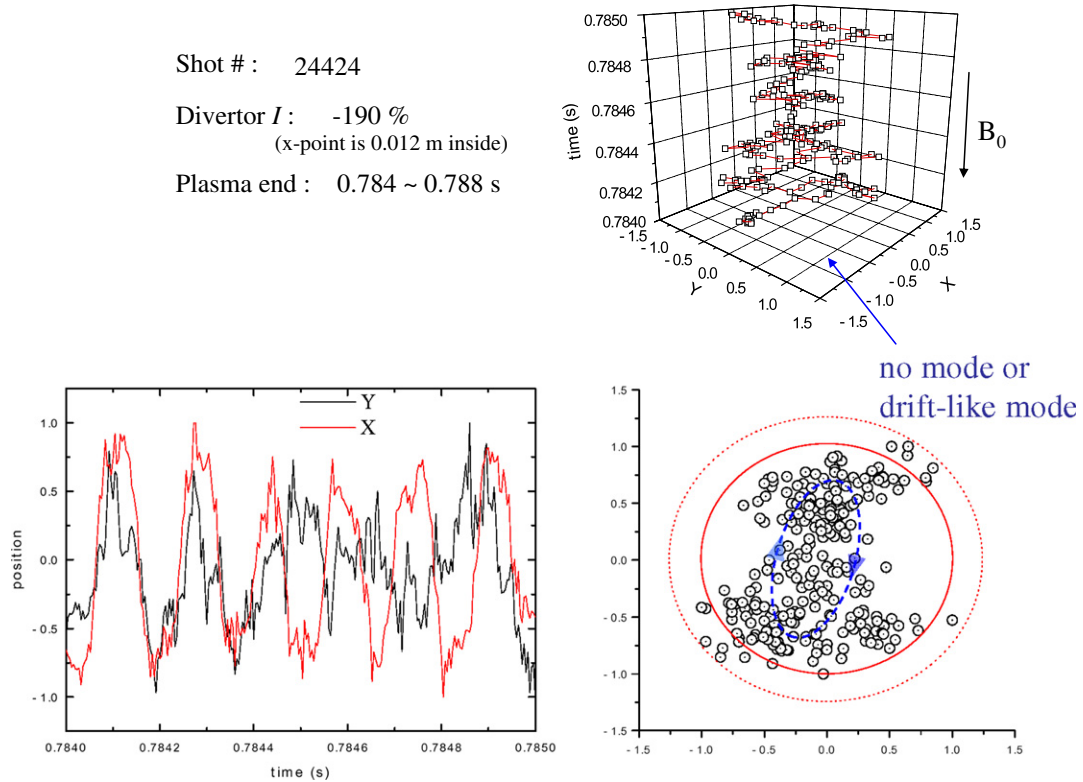


Figure 13. Shot 24424 with a divertor current of -190% . The frame identifications are the same as in figure 12. The motion is unclear but appears to be a mixture of flute- and drift-like modes.

Shot #: 24497_E1564_3

Divertor I : -195 %

Plasma end: 0.583 ~ 0.585 [s]

Divertor start time: ~6.05 [s]

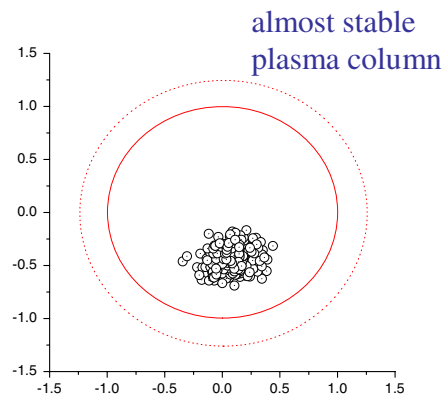
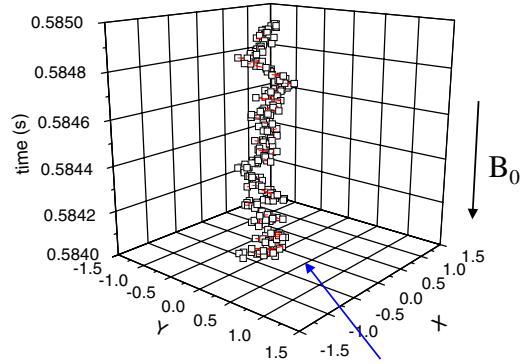
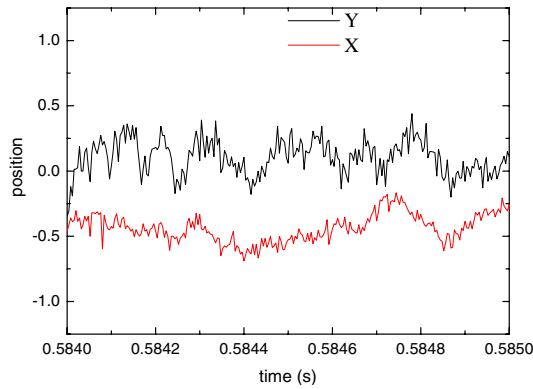


Figure 14. Shot 24497 with a divertor current of -195% . The frame identifications are the same as in figures 12 and 13. The plasma appears to be stable.

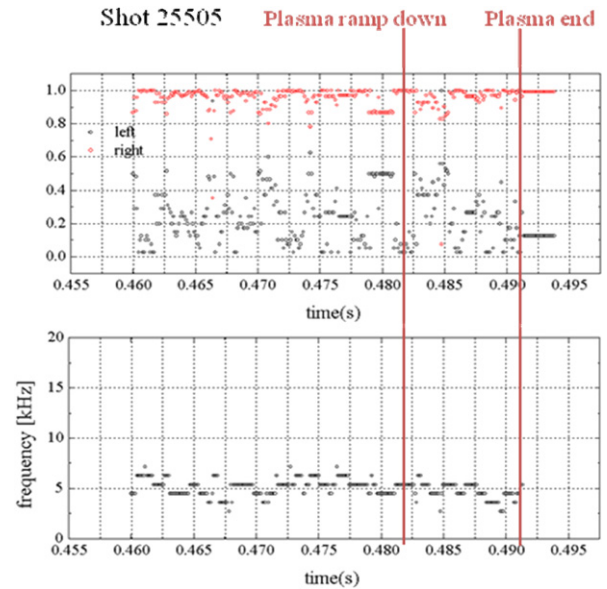
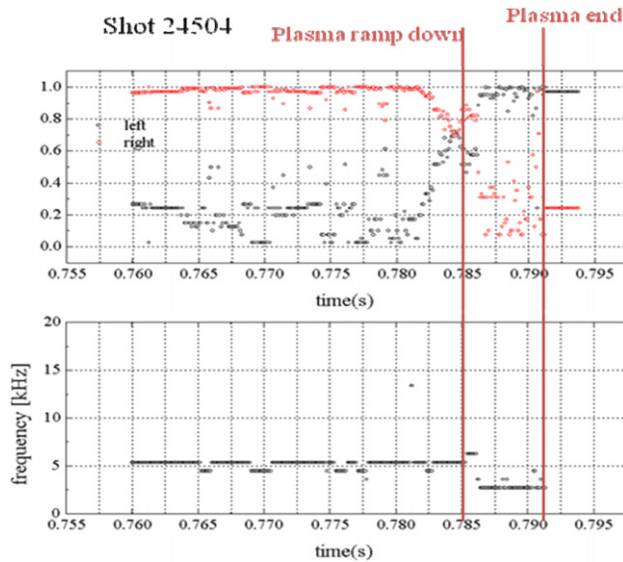


Figure 15. Top: right and left propagating components of the plasma for a shot with the x-point outside the vacuum vessel. The initial drift mode becomes a flute mode. Bottom: the frequency of the drift mode is ~ 5 kHz and the frequency of the flute mode is ~ 3 kHz.

Figure 16. Right- and left-rotating components for a shot with the divertor current high enough to provide stabilization. The right-rotating component in the top part of the figure remains dominant to the end of the shot. The bottom part of the figure shows that the frequency remains the same as the drift frequency near 5 kHz.

in the early part of the shot with a frequency near 5 kHz is replaced by a flute mode at the end with a frequency near 3 kHz.

Figure 16 shows this same analysis for a shot with a divertor current of -200% where the divertor stabilization

is sufficient. In the top part of the figure the right-rotating component remains dominant and never changes into the left-rotating component. In the bottom part of the figure the frequency remains the same as the drift frequency.

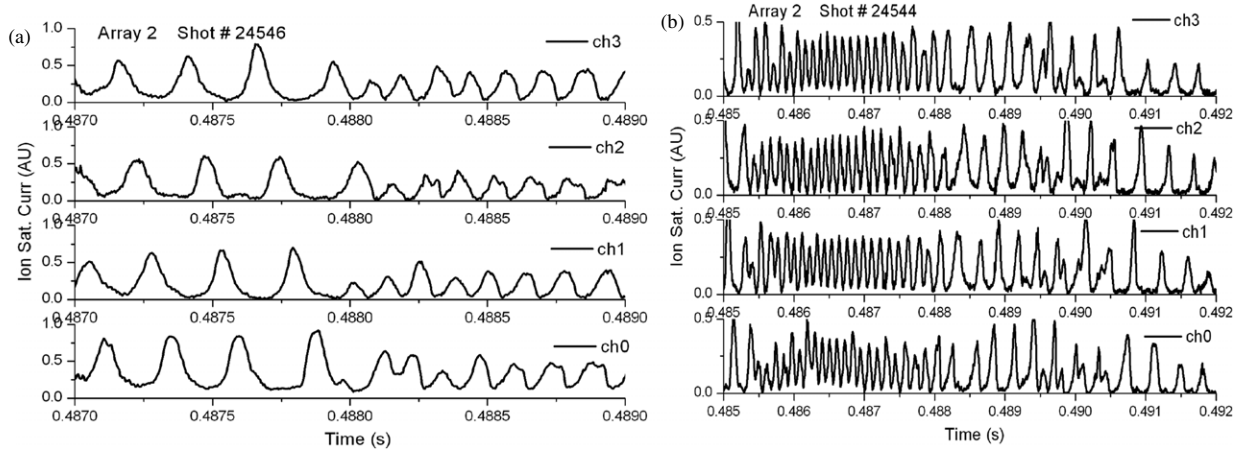


Figure 17. Comparison of shots with and without the KS plasma. (a) With the KS plasma present the instability duration is about 1.5 ms. (b) Without the KS plasma, the instability duration is about 6 ms.

7. KS design and first KS plasma measurements

We reported earlier [23] on the experiments aimed at creating KS plasmas in the end region of Hanbit. We produced electron cyclotron resonant heated (ECRH) hydrogen plasmas in a tailored expanding field region in the cusp tank using a 5 kW 2.45 GHz magnetron. We reported that the temperature of the ECRH plasma was 15–30 eV which was as much as a factor of 10 higher than the plasma escaping from the central cell and that the density was also higher. However, we were unable to attempt stabilization experiments since the ion cyclotron resonant heating (ICRH) central cell plasmas created at that time did not exhibit the flute-like $m = -1$ MHD instability. In the ensuing period we found conditions to create this instability. The same magnetron was also used for the experiments reported here. It was not possible to control the turn on and turn off times precisely so magnetron was turned on before the ICRH plasma was created and turned off after the ICRH plasma terminated.

8. Most recent KS experiments

As mentioned earlier in regard to the divertor experiments, for 200 kW ICRH plasmas, we have found conditions so that there is normally a disruptive $m = -1$ flute-like MHD instability in the low density plasma at the end of the discharge as the ICRH power ramps down. If the ICRH trips suddenly and a smooth ramp down to zero power is not present, there is no such $m = -1$ instability [24]. Under these conditions, an $m = +1$ drift-wave-like instability is normally seen. We have been careful not to include in the analysis any shot which does not have a smooth ramp down to zero power. In addition, we were able to operate at slightly higher coil currents (135%) compared with the earlier Hanbit experiment [18] coil currents (130%). We also turned off the anchor coil for these experiments so it would not influence the stability.

The signals from an azimuthal array probe mounted on a limiter are shown in figure 17(a) for shot 24546. Figure 17(b) shows shot 24544 with this same instability but with a shorter duration. Repeated experiments have shown that the instability is almost always shorter with the KS plasma and as a result,

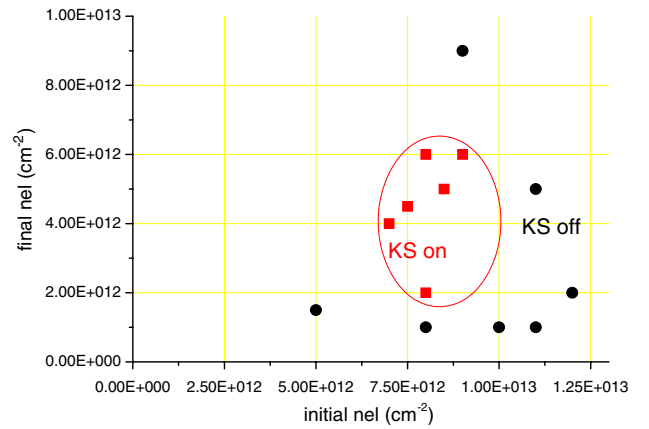


Figure 18. Range of line density over which the $m = -1$ instability was created during ramp down of ICRH power from 200 kW. The range of line density and the duration of the signal are both reduced for plasmas with the KS plasma present in the cusp.

the range of line density over which the instability occurs is smaller.

Figure 18 shows the line density range over which we have observed the $m = -1$ instability at the ICRH ramp down for 200 kW plasmas. The circles are for the normal ramp down without the KS plasma in the cusp. The squares inside the ellipse show that the KS plasma reduces the line density range over which the $m = -1$ instability is present. The corresponding average density ranges are 3×10^{16} to $3 \times 10^{17} \text{ m}^{-3}$ for the plasmas without the KS plasma on and 5×10^{16} to $2 \times 10^{17} \text{ m}^{-3}$ with the KS plasmas.

A search was made for high density ($\leq 1 \times 10^{18} \text{ m}^{-3}$) plasmas that would disrupt with an $m = -1$ instability. By carefully controlling the magnetic field and gas puffing, a condition was found at an ICRH power of 80 kW which normally terminates during the ICRH pulse due to this instability. As before, we carefully excluded all shots which had an ICRH trip because such shots never display the $m = -1$ signature.

Figure 19 shows the signals from an azimuthal array of probes on one of the limiters for shot 25201 with the KS

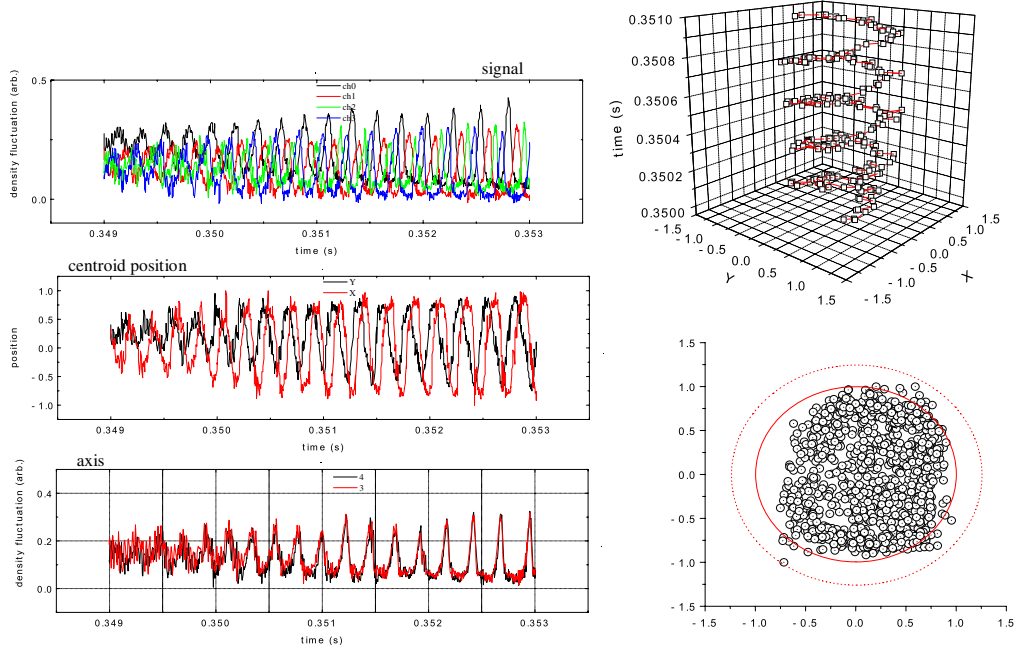


Figure 19. Flute-like mode for a shot without the KS on. A clear $m = -1$ flute-like mode is shown.

plasma off. In this shot the ICRH power remained on until long after the plasma terminated. The instability duration was about 5 ms for shot 2501. Although the frequency varied somewhat during the instability, the phase relation between the probes was maintained for the entire duration of the instability. As with figures 12–14, the top right graph shows a 2D representation of the centroid versus time and the bottom right graph shows an X – Y plot of the centroid as viewed from above. On the left side at the top are the four probe signals versus time, in the middle left the X and Y values of the centroid position are plotted versus time and on the bottom left two probe signals from two probes at the same azimuth on each end of the device are plotted versus time to show coherence. We conclude that $k_{\parallel}$ is approximately zero, i.e. it is an $m = -1$ flute.

Figure 20 shows the signals from an azimuthal array of probes on one of the limiters for shot 25202 with the KS on. In this shot the ICRH power remained on until long after the plasma terminated. The instability duration was about 17 ms for shot 25202. Although the frequency varied somewhat during the instability, the phase relation between the probes was maintained the entire duration of the instability. As with figures 17–19, the top right graph shows a 2D representation of the centroid versus time and the bottom right graph shows an X – Y plot of the centroid as viewed from above. On the left side at the top are the four probe signals versus time, in the middle left the X and Y values of the centroid position are plotted versus time and on the bottom left two probe signals from two probes at the same azimuth on each end of the device are plotted versus time to show coherence. We conclude that $k_{\parallel}$ is approximately 0, i.e. it is an $m = -1$ flute. While the character of the pattern has changed somewhat, a clear $m = -1$ signal is still present. The rectangular appearance is due to the limited number of probes making contact with the plasma whose radius has shrunk. Each probe produces a signal when the $m = -1$ perturbation makes contact with it. If there were

more probes the appearance would be more circular. However, even with just four probes it is clear that this is a CCW rotating flute-like mode.

Figures 21(a) and (b) show the signals from an azimuthal array of probes on one of the limiters for the same two shots as for figures 19 and 20. Although the frequency varies somewhat during the instability, the phase relationship between the probe signals is maintained for the entire instability duration. A precise measurement of the duration is difficult due to the noisy nature of the signal but it is clear that the KS plasma shot $m = -1$ signal duration is much longer than the duration for the shot without the KS plasma. If the hydrogen gas pressure in the cusp region was raised sufficiently high, both the normal shots as well as the KS shots showed a reduction in the instability duration. That is to say, high gas pressure in the cusp region decreased the instability duration for all of the plasmas. However, up to the limitations of the experiment, in no case did we find a condition without an $m = -1$ instability or with the KS plasma having a shorter instability period than the normal plasma.

A similar analysis for the period of time when the drift-like mode is active shows that the probe signals display $k_{\parallel} \sim 0$. The data are too noisy to make an exact determination. This problem was analysed by Kim and Jhang [25] for the Hanbit conditions from a two fluid approach. According to figure 1(a) of [25], as the parallel wave number increases the mode changes from an interchange type to a drift type and eventually becomes an ion acoustic mode. However, the wave numbers are still small for the drift mode and cannot be experimentally distinguished from zero due to noise. Experimentally, we have found that drift and interchange modes can be identified by the direction of the phase velocity. The drift mode frequency is similar but not identical to the flute mode as discussed in section 5.

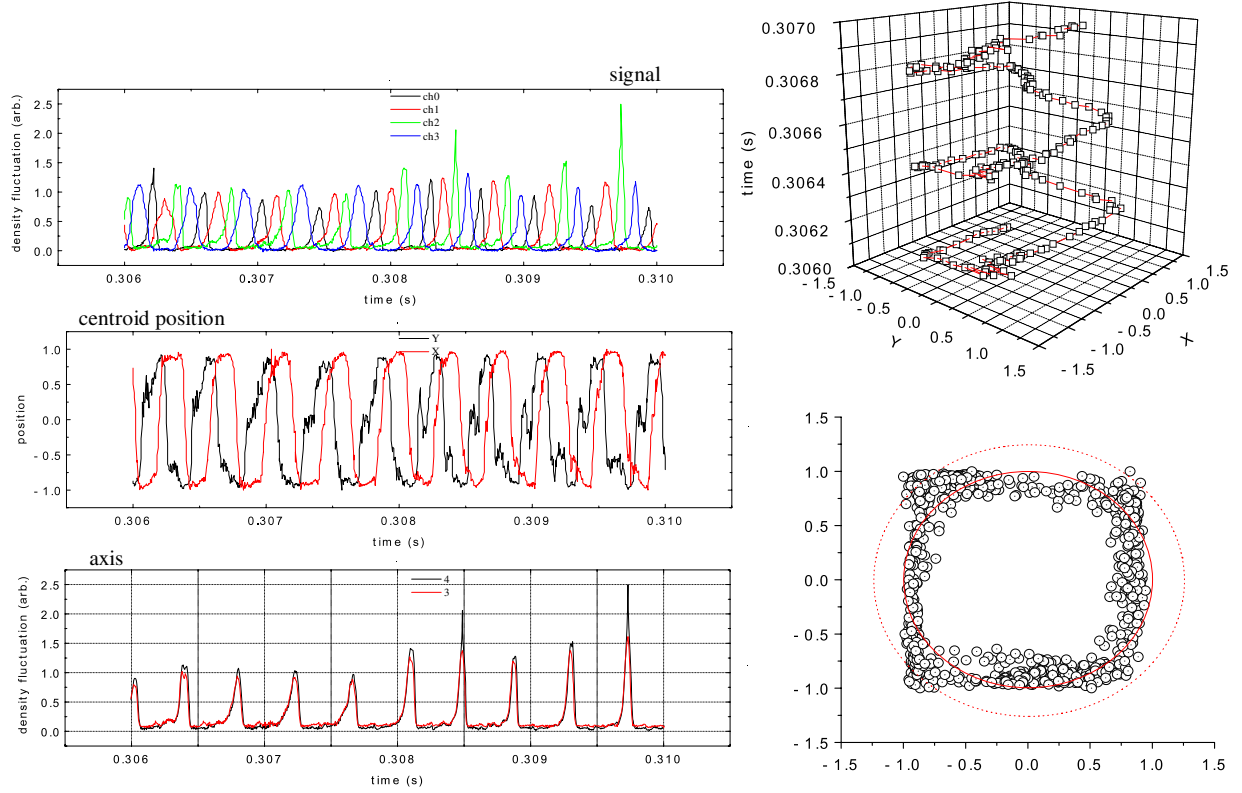


Figure 20. Flute-like mode for a shot without the KS on. A clear $m = -1$ flute-like mode is shown.

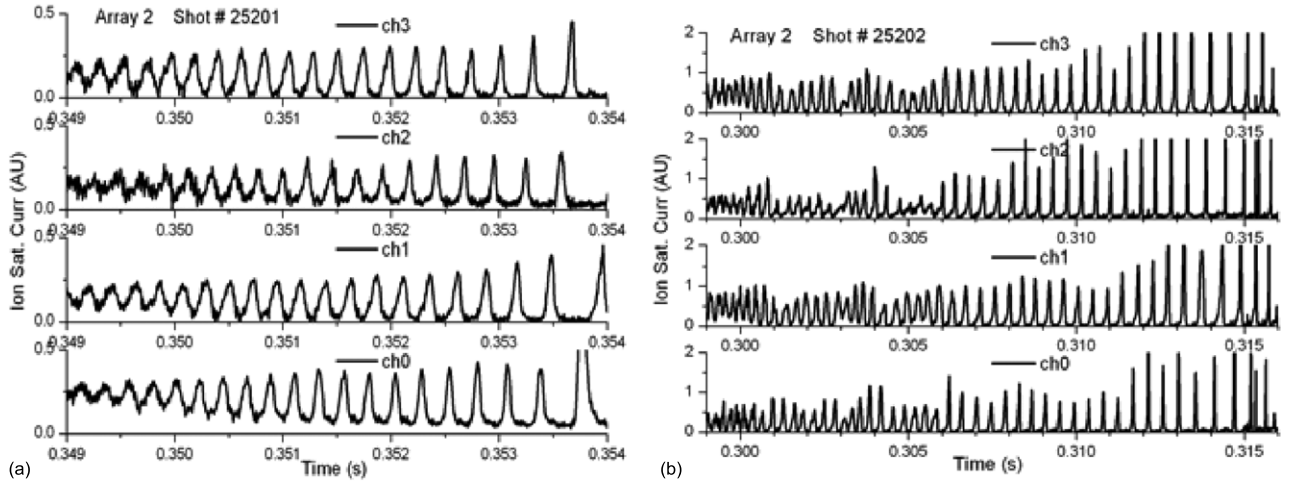


Figure 21. (a) Shot 25201 without KS plasma. (b) Shot 25202 with KS plasma. The $m = -1$ signal duration is longer for shot 25202 with the KS on.

9. Discussion

We do not have the facilities to test the concept according to the original plan of Post using low energy ion beams that stagnate in the cusp. Because there can be other factors over which we have no control, we cannot make a judgment on the validity of the concept of the KS, but only that our particular application is ineffective. Most importantly, we were not able to measure the parallel plasma pressure, $p_{\parallel}$, or the perpendicular plasma pressure, $p_{\perp}$, of the KS plasma on and near the outermost field line to determine whether the stability integral I was greater than zero. We were also unable to install an emitting probe and

a Mach probe in the cusp to measure the space potential and the plasma flow velocity, respectively. As a result, we do not know whether those parameters of the KS plasmas have any influence on the results. The Hanbit device has been dismantled and no further experiments are possible.

10. Summary and conclusions

The Hanbit device has been involved in a series of experiments on stabilization of the MHD flute-like mode. Divertors were used previously in experiments on the TARA mirror device [4, 5] and the HIEI mirror device [6, 7]. The chosen

configuration used just one divertor coil in one end of Hanbit and produced a left-right asymmetry in the magnetic field. The plasma was difficult to establish and maintain with this configuration. However, for sufficiently high currents this divertor was able to stabilize the $m = -1$ MHD flute-like instability at the end of a normal discharge. The stability condition is in good agreement with that expected from the calculated fields. KS experiments with low density plasmas showed a small decrease in the instability duration with the KS plasma but experiments with high density ICRH plasma showed that the $m = -1$ instability duration was increased by the KS plasma while an $m = +1$ instability earlier in the discharge was reduced.

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